Downforce Optimization of an Airfoil in Ground Effect

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Application and model. We will optimize a single-element racing-car front-wing section to maximize downforce under drag and geometric limits. The starting section is NACA 0012 with chord $c = 0.50\,\mathrm{m}$, initial minimum ground clearance $0.05\,\mathrm{m}$, speed $200\,\mathrm{km/h}$, and fixed incidence -3° (nose down). At $288.15\,\mathrm{K}$ and $101325\,\mathrm{Pa}$, $M \approx 0.163$ and $Re_c \approx 1.90 \times 10^6$. We use the open-source MACH-Aero framework: ADflow solves steady, fully turbulent Spalart–Allmaras RANS on a body-fitted O-grid overlapping a background grid. The quasi-2D mesh has one spanwise cell and 88,548 cells before blanking. The ground is an ideal slip/symmetry plane, not a moving no-slip road; finite-span and vehicle effects are excluded.

Inputs, outputs, cost, and derivatives. Inputs are section-shape variables and prescribed operating conditions; outputs are C_L, C_D, surface pressure, thickness, and clearance. The converged baseline gives $C_L = -1.1518$, $C_D = 0.02296$, and maximum $y^+ = 0.433$. A fresh flow evaluation took approximately $150\,\mathrm{s}$ on four local MPI processes; adjoint cost is not yet measured. We will use pyGeo free-form deformation (FFD), IDWarp mesh deformation, and ADflow discrete-adjoint aerodynamic derivatives, checking selected derivatives against finite differences. Optimization solves will initially target flow residual reduction 10^{-10} and adjoint reduction 10^{-12}, with force and gradient sensitivity checks. Mesh independence remains to be established.

Optimization problem. Let $x \in \mathbb{R}^{20}$ be vertical FFD displacements at ten upper/lower chordwise stations, coupled between spanwise planes; endpoint constraints preserve chord and incidence. Let u denote the flow state. With upward-positive C_L, minimizing C_L maximizes downforce. Our proposed initial design limits define

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\begin{aligned}
\min_{x,u} \quad & C_L(x,u) \\
\text{subject to} \quad & r(u,x) = 0, \qquad Ex = 0, \\
& C_D(x,u) \le C_{D,\max} = C_{D,0} \approx 0.02296, \\
& t_j(x) \ge 0.5\, t_{j,0}, \qquad j = 1,\dots,10, \\
& 0.05\,\mathrm{m} \le y_k(x) \le 0.20\,\mathrm{m}, \qquad k = 1,\dots,N_s, \\
& -0.01\,\mathrm{m} \le x_i \le 0.01\,\mathrm{m}, \qquad i = 1,\dots,20.
\end{aligned}
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Here r is the discrete RANS residual, $Ex = 0$ enforces endpoint restrictions, and y_k is a deformed surface-point height above the ground. Thickness t_j is measured along fixed chord-normal directions at $s_j/c = 0.05, 0.15, \dots, 0.95$; subscript 0 denotes the baseline. Geometry validity and overset connectivity will be checked for every candidate. This is a continuous, constrained, nonconvex, PDE-constrained nonlinear program, treated in reduced space by solving for $u(x)$.

Problem sequence and software. P0: establish mesh sensitivity, tighter flow convergence, and derivative accuracy. P1: map incidence and clearance effects with low-dimensional sweeps. **P2 (minimum deliverable):** optimize the full single-element shape at the stated conditions and compare against the baseline on a refined mesh. P3: sweep the drag cap to trace the downforce–drag trade-off. Stretch goals are sensitivity to the packaging envelope and clearance, an IPOPT interior-point comparison, and a two-element wing. The primary optimizer is SLSQP (sequential quadratic programming) through pyOptSparse; a custom optimizer is outside the minimum scope.

Expected learning. We will determine how much downforce shape changes can add at fixed drag, whether drag, thickness, clearance, or the upper envelope limits the design, and how pressure recovery and separation explain the result. We will assess robustness to mesh refinement and distinguish improvements within the slip-ground model from predictions for a complete race car.